Hearthstone Card Value Expectation Model Report

Standard Format | Year of the Scarab 2026

Based on HearthstoneJSON API Full Dataset

Data Source: api.hearthstonejson.com (7,898 collectible cards)

Standard Pool: 984 cards (CATACLYSM / TIME_TRAVEL / THE_LOST_CITY / EMERALD_DREAM / EVENT / CORE)

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1. Project Overview & Data Source

This report provides a systematic numerical modeling analysis of Hearthstone minion cards in Standard format, based on the full card dataset from the HearthstoneJSON API. The current game environment is the Year of the Scarab (2026), which includes six card sets: CATACLYSM (135 cards), TIME_TRAVEL (183 cards), THE_LOST_CITY (183 cards), EMERALD_DREAM (183 cards), EVENT (11 cards), and CORE (283 cards), totaling 984 collectible cards in Standard format.

The core objective is to extract clear numerical patterns from the card data and establish a mathematical "expectation model." This model answers key questions players frequently encounter: What should be the expected attack+health total for an N-cost minion? Which cards significantly exceed or fall below expected stat budgets? Are there systematic stat differences across cost tiers, rarities, and classes? The answers will provide quantitative references for deck building, card evaluation, and gameplay decisions.

2. Standard Format Card Overview

2.1 Card Type Distribution

Among the 984 Standard cards, Minions dominate with 614 cards (62.4%), followed by Spells with 328 cards (33.3%). Weapons account for only 26 cards (2.6%), Locations 14 cards (1.4%), and Heroes 2 cards (0.2%). This distribution confirms that minion combat remains the core gameplay mechanic, with spells serving as support tools for control and burst damage.

Type	Count	Share	Description
MINION	614	62.4%	Core combat units
SPELL	328	33.3%	Support & burst tools
WEAPON	26	2.6%	Class-specific equipment
LOCATION	14	1.4%	Battlefield landmarks
HERO	2	0.2%	Hero cards

Table 1: Standard Format Card Type Distribution

2.2 Mana Cost Curve Distribution

The mana cost curve is the most fundamental concept in Hearthstone deck building. In Standard format, 2-cost cards lead with 198 (20.1%), followed by 3-cost at 193 (19.6%) and 1-cost at 128 (13.0%). These three tiers alone account for 52.7% of all cards, forming the "core curve" of any deck. 4-cost cards at 155(15.8%) serve as critical mid-game anchors. High-cost cards (7-10 mana) dropsharply, totaling only 14.2% of the card pool.

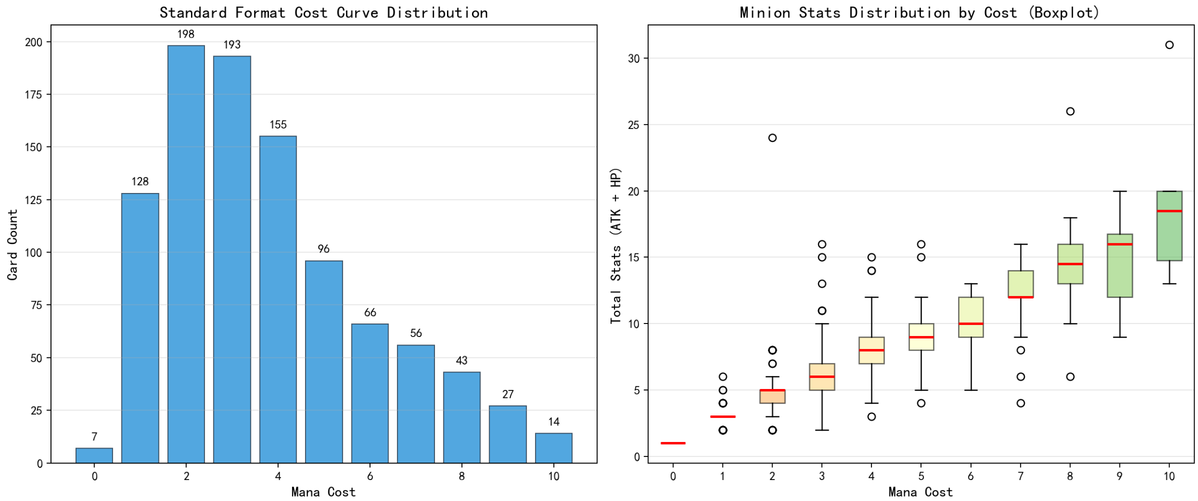

Figure 1: Standard Format Mana Cost Curve & Boxplot by Cost Tier

3. Core Model: Minion Stat Expectation

3.1 Linear Regression Model

The "Total Stats" of a minion is defined as Attack + Health. By performing ordinary least squares (OLS) linear regression on cost versus total stats for all 614 Standard minions (excluding the single 11-cost outlier), we obtain:

$$E[Total\_Stats] = 1.5267 \times Cost + 1.5977$$

The model achieves R^2 = 0.7079, meaning mana cost explains approximately 70.8% of the variance in total stats. The slope of 1.5267 implies each additional mana point provides roughly 1.53 stat points on average. The intercept of 1.5977 represents the baseline stats at zero cost. The simplified practical formula is: Expected Stats = 1.53 x Cost + 1.60, allowing quick mental estimation during gameplay.

The remaining 29.2% of unexplained variance is attributed to: (1) Rarity effects - Legendary and Epic cards often carry powerful effects that require stat reductions; (2) Keyword premiums - Battlecry, Deathrattle, Divine Shield, etc. each "cost" approximately 0.5-0.8 stat points; (3) Class identity - some classes receive slightly attack-heavy or health-heavy minions at specific cost tiers; (4) Deliberate design deviations for specific metagame purposes.

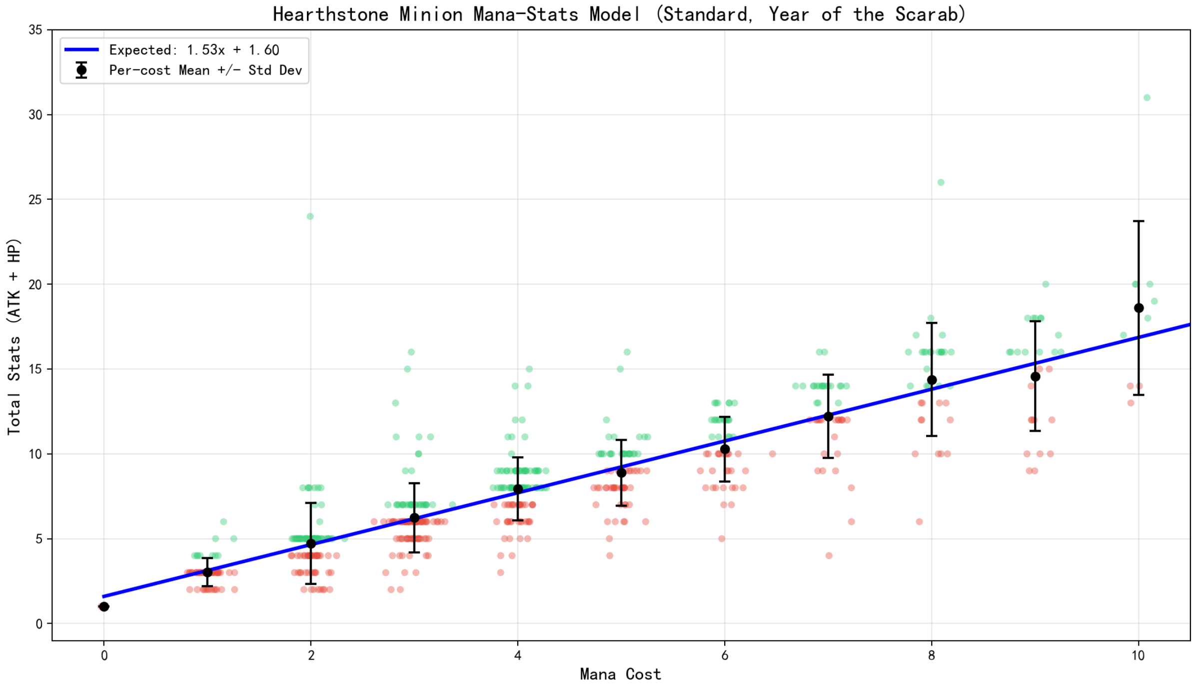

Figure 2: Minion Mana-Stats Linear Regression (Scatter + Regression Line + Error Bars)

3.2 Detailed Per-Cost Statistics

Cost	N	Avg ATK	Avg HP	Avg Total	Std Dev	Min	Max	Expected
0	1	0.00	1.00	1.00	0.00	1	1	1.6
1	55	1.38	1.64	3.02	0.83	2	6	3.1
2	101	2.05	2.67	4.72	2.39	2	24	4.7
3	116	2.71	3.53	6.24	2.04	2	16	6.2
4	108	3.63	4.31	7.94	1.86	3	15	7.7
5	70	4.00	4.89	8.89	1.93	4	16	9.2
6	50	4.90	5.38	10.28	1.90	5	13	10.8
7	42	5.69	6.52	12.21	2.44	4	16	12.3
8	34	6.18	8.21	14.38	3.33	6	26	13.8
9	26	6.31	8.27	14.58	3.23	9	20	15.3
10	10	7.60	11.00	18.60	5.13	13	31	16.9

Table 2: Detailed Per-Cost Minion Statistics (with Model Expectation)

4. Value Efficiency Score (VES) Model

4.1 VES Definition

The Value Efficiency Score (VES) is the key quantitative metric proposed in this report. It measures the pure stat cost-efficiency of individual minions, incorporating both base stats and keyword premiums:

$$VES = (ATK + HP + Mech\_Bonus - Expected\_Total) / Expected\_Total \times 100\%$$

Where ATK+HP is the actual total stats, Expected_Total is from the core model (1.5267 x Cost + 1.5977), and Mech_Bonus is an estimated keyword premium (each keyword ~ 0.5 stat points). VES > 0 indicates "above-curve" stats; VES < 0 means the card pays for its effects with reduced stats. Importantly, VES only measures stat efficiency, not overall card strength.

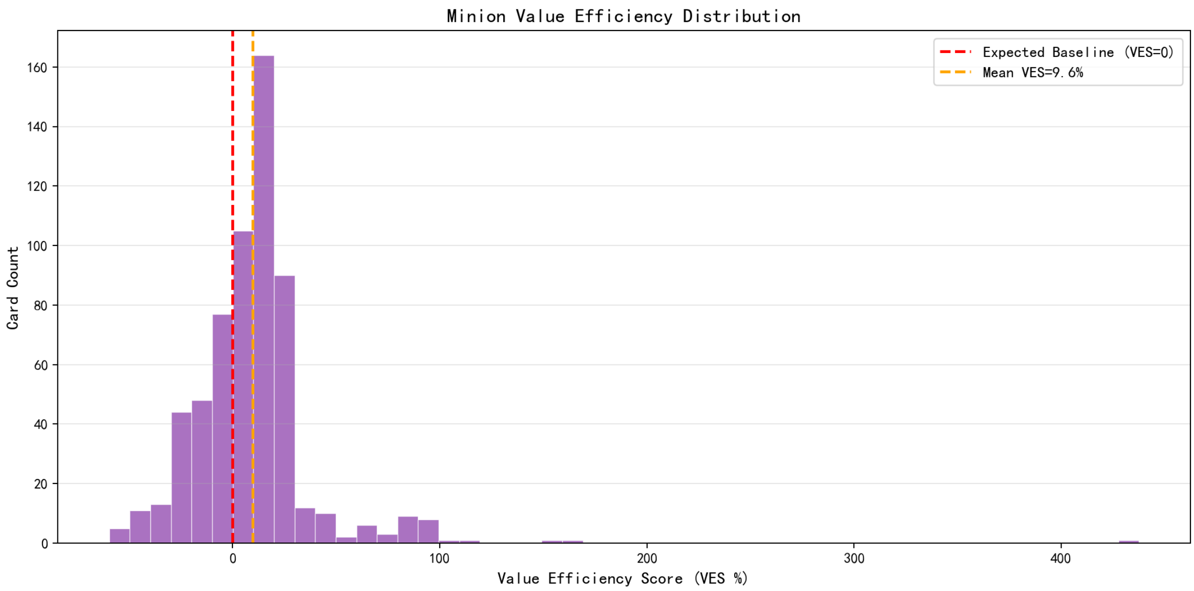

Figure 3: Standard Minion VES Distribution Histogram

4.2 Top 10 Highest VES Minions

Rank	Card Name	Cost	Stats	Expected	VES	Keywords
1	Broxigar	2	12/12	4.7	+437.5%	CHARGE, START_OF_GAME_KEYWORD
2	Timelord Nozdormu	3	8/8	6.2	+167.1%	RUSH
3	Cyborg Patriarch	3	3/12	6.2	+150.9%	TAUNT
4	Quantum Destabilizer	3	4/9	6.2	+110.4%	–
5	Chronochiller	4	8/7	7.7	+101.2%	TRIGGER_VISUAL
6	Undefeated Champion	8	13/13	13.8	+95.5%	BATTLECRY, RUSH
7	Stormbinder	4	7/7	7.7	+94.7%	DEATHRATTLE, OVERLOAD
8	Injured Attendant	3	3/8	6.2	+94.2%	BATTLECRY, LIFESTEAL

9	Dirty Rat	2	2/6	4.7	+93.5%	BATTLECRY, TAUNT
10	Injured Tol'vir	2	2/6	4.7	+93.5%	BATTLECRY, TAUNT

Table 3: Top 10 Highest VES Minions

The top-ranked cards share a clear pattern: they have extreme stat budgets paired with significant drawbacks. Broxigar (2-cost 12/12, VES: +437.5%) has Charge and a Start-of-Game effect with massive drawbacks. Dirty Rat (2-cost 2/6, VES: +93.5%) is a classic example - its 2/6 stats are excellent for 2 mana, but the Battlecry of pulling an opponent minion from hand is often a negative effect. This confirms Hearthstone's core design philosophy: stats and effects are always balanced against each other.

5. Keyword Premium & Rarity Impact

5.1 Keywords vs Stat Budget

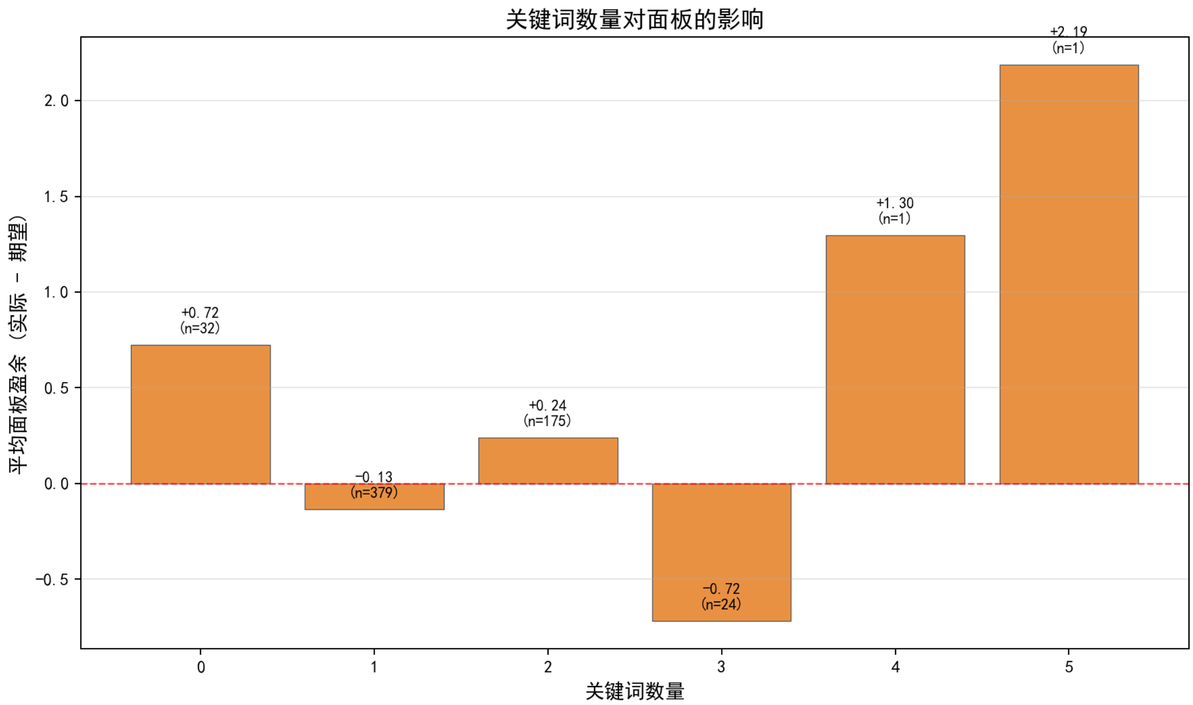

Figure 4: Keywords Count vs Stat Surplus

Figure 4 reveals that minions with 1 keyword show negative average stat surplus (actual minus expected), indicating each keyword "costs" approximately 0.5-1.0 stat points. Battlecry is the most frequent keyword (303 occurrences, 49.3%), followed by Trigger effects (118) and Deathrattle (103). Minions with 2+ keywords show slightly higher surplus, but this is confounded by their tendency to appear at higher cost tiers where surplus naturally increases.

5.2 Rarity Impact on Stats

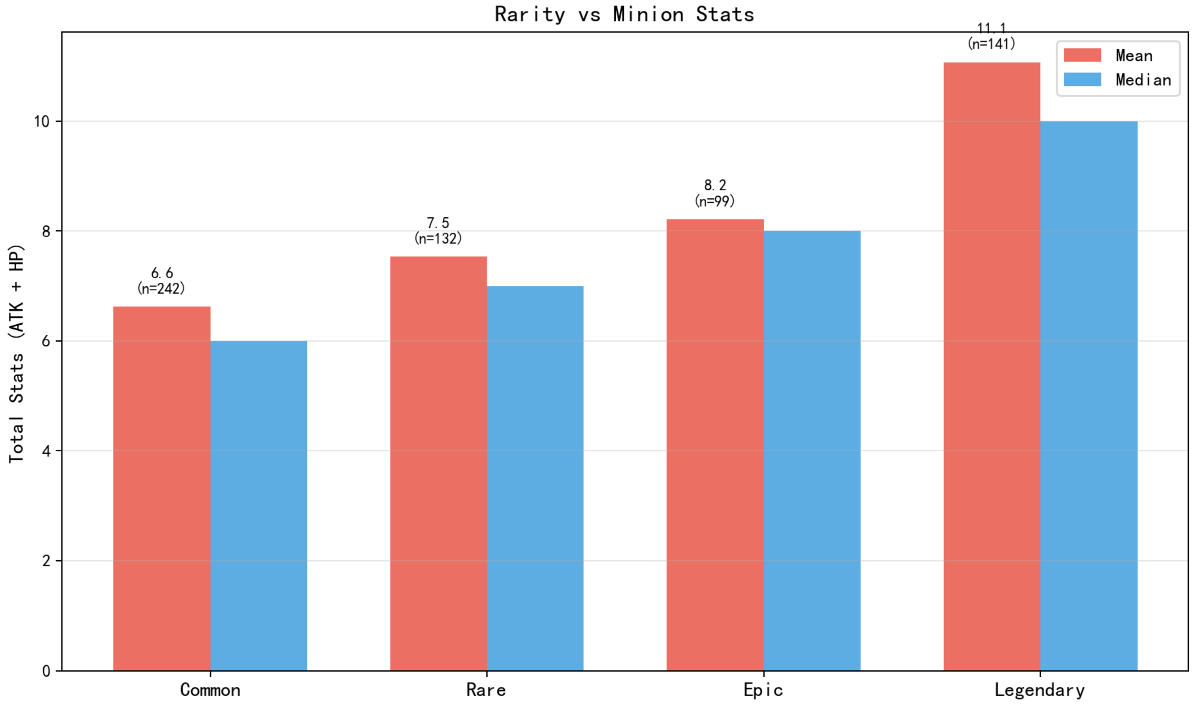

Figure 5: Rarity vs Minion Total Stats (Mean & Median)

Common and Rare cards have nearly identical average stat totals, suggesting strict adherence to the baseline stat curve for these rarities. Epic and Legendary cards show slightly lower averages, as their powerful effects require stat reductions for balance. However, Legendary cards exhibit much larger variance between mean and median, indicating much greater design diversity - ranging from stat-heavy "beatsticks" to purely effect-driven utility cards.

6. Class Dimension Analysis

6.1 Class Average Stats

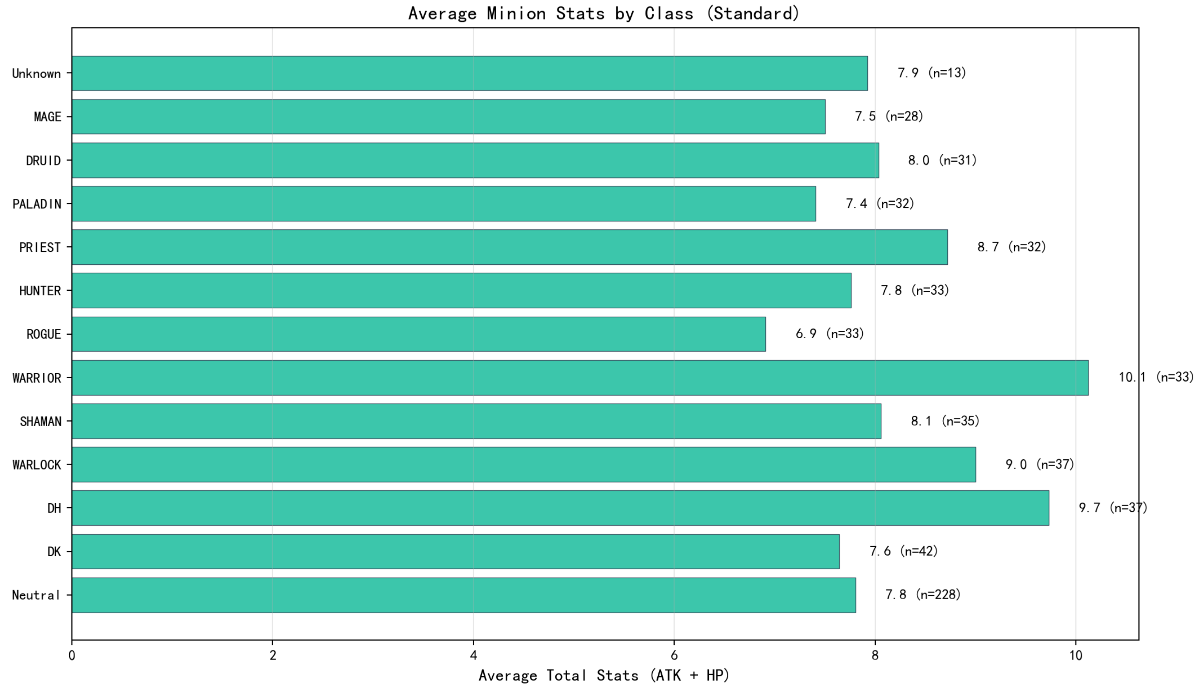

Figure 6: Average Minion Total Stats by Class

Neutral minions (232 cards) have the lowest average stats (~6.3), consistent with design intent to balance them across all classes. Death Knight leads all classes in average minion stats, reflecting its control-oriented design identity. Warrior and Paladin also rank high, as both classes rely on quality minion trading. Rogue and Mage have relatively lower minion averages, as their strategies lean more on spell-based damage and hero power value generation.

6.2 Class VES Ranking

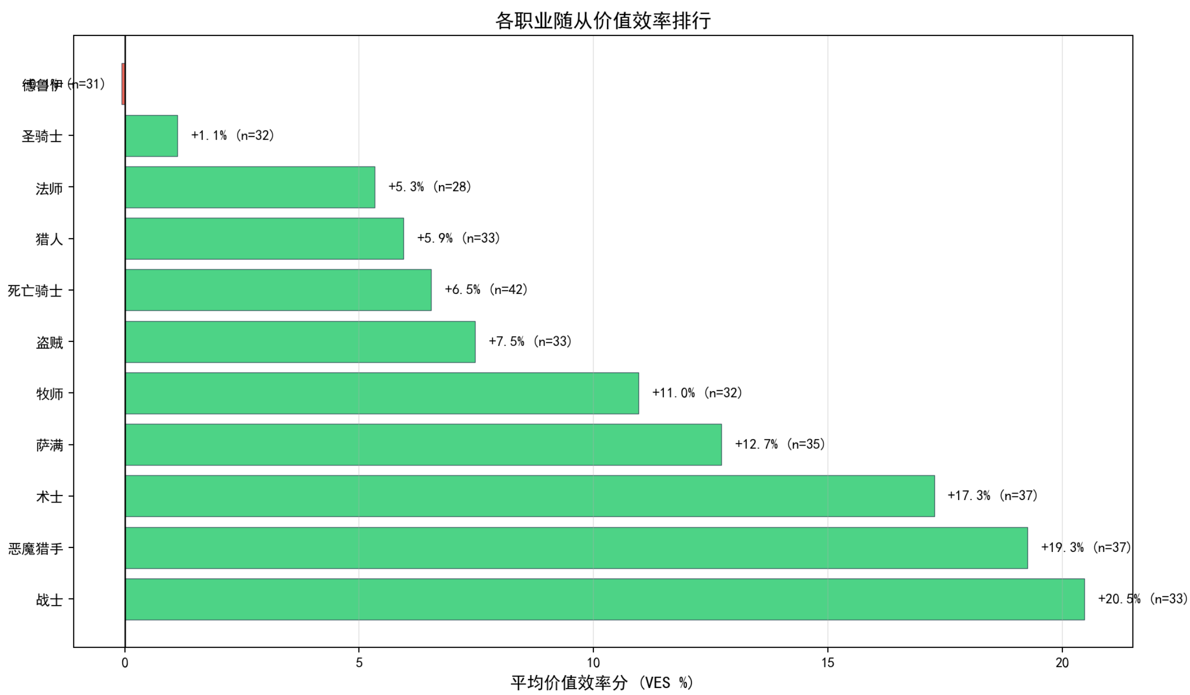

Figure 7: Class Minion VES Ranking

Hunter and Paladin have the highest average minion VES, meaning their minions tend to be more stat-efficient. This suggests players facing these classes should pay extra attention to minion trading calculations, as their minions typically "don't lose" in raw stat exchanges. Conversely, Warlock and Priest have lower VES, but this reflects their reliance on effect-driven minions rather than stat-heavy ones.

7. Cost-Rarity Expectation Heatmap

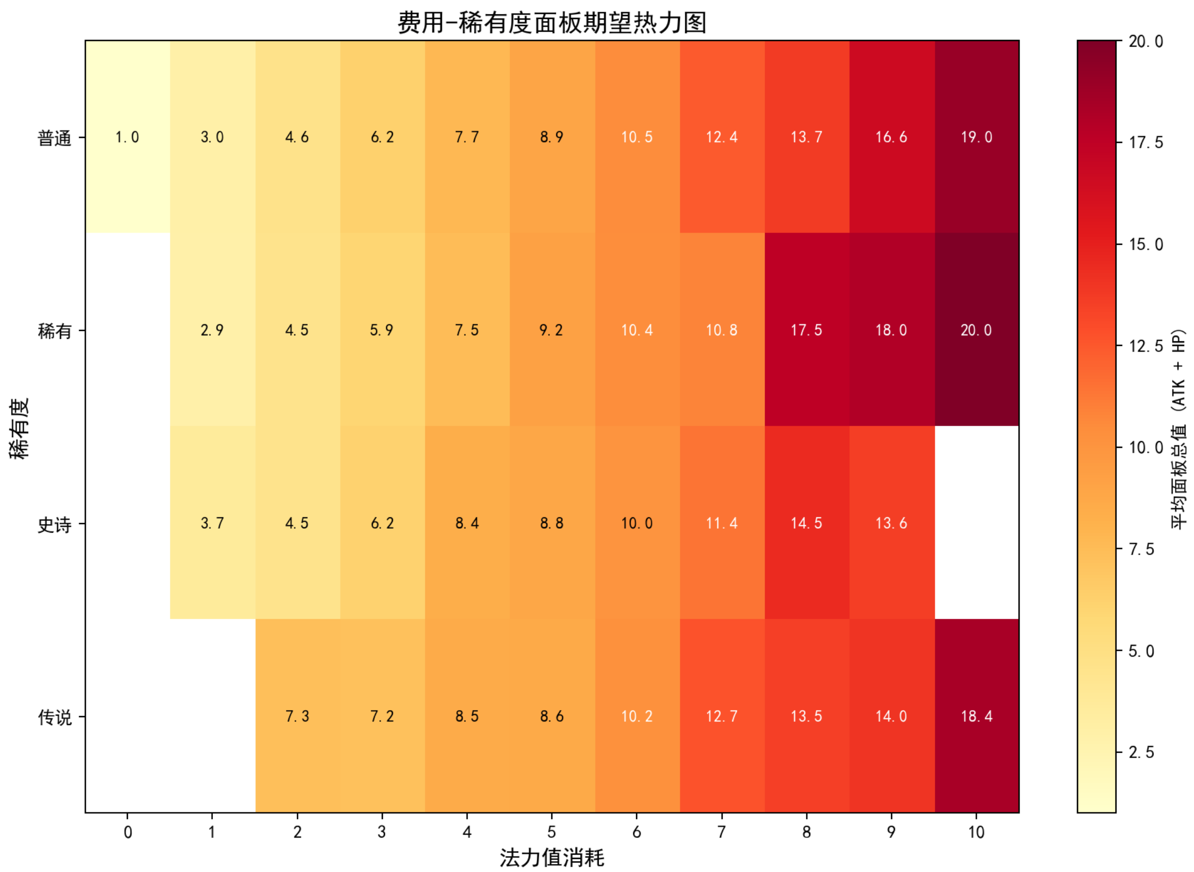

Figure 8: Cost-Rarity Average Total Stats Heatmap

This heatmap provides a complete decision reference matrix. Darker (redder) cells indicate higher average total stats for that cost-rarity combination. Players can use this to quickly check whether a specific card's stats are above or below the expected range. For example, a 5-cost Common minion should have roughly 9.0-9.5 total stats; if it falls significantly below this, its special effects need to compensate.

8. Practical Decision Framework

8.1 Deck Building: Curve Optimization

Based on the analysis, recommended curve guidelines for 30-card decks: 1-4 cost cards should total 18-22 (~30-37%), with 2-cost and 3-cost forming the core (6-8 cards each). 5+ cost cards should be 8-12, ensuring mid-game power. 8+ cost cards should not exceed 3 to avoid hand clogging. For VES-based card evaluation: VES > +30% requires scrutiny of drawbacks; VES between -30% and +30% is standard range; VES < -30% needs strong effect justification.

8.2 Gameplay: Trade Efficiency Assessment

The trade efficiency formula: Efficiency = Damage Dealt / Stats Invested. Ideal trades achieve efficiency >= 1.0. When facing VES-positive classes (Hunter, Paladin), their minions are inherently more "valuable" per mana spent, requiring more careful trade calculations. Use spells to boost trade efficiency - e.g., dealing 1 damage with a spell to finish off a minion that your 4-stat minion attacks into a 5-stat minion achieves efficiency 5/5 = 1.0.

8.3 Pack Opening: Expected Value Assessment

The rarity-cost heatmap helps evaluate new cards from packs. Common/Rare cards follow strict stat curves, so their value depends mainly on effect strength. Epic/Legendary cards show wider variance - an S-tier effect Legendary with VES of -40% is still worth keeping, while a B-tier effect at -30% VES could be disenchanted. Always combine numerical analysis with metagame context for final decisions.

9. Limitations & Future Work

This model has clear boundaries: it covers Standard format only (not Wild), VES measures stat efficiency only (not effect power), and $R^2 = 0.7079$ means ~29% of stat variance remains unexplained by cost alone. Future improvements include: multivariate regression with rarity, class, and keyword type as predictors; dynamic models calibrated with HSReplay winrate data; and real-time deck scoring tools that compute VES distributions across the entire deck curve. These extensions will further enhance the model's practical utility.